Q1 - 2024 (04 Apr Shift 1)

A hydrogen atom changes its state from $n=3$ to $n=2$. Due to recoil, the percentage change in the wave length of emitted light is approximately 1×10^{-n}. The value of n is_____.

[Given $\mathrm{Rhc}=13.6 \mathrm{eV}, \mathrm{hc}=1242 \mathrm{eVnm}, \mathrm{h}=6.6 \times 10^{-34}$ J s mass of the hydrogenatom $=1.6 \times 10^{-27}$ kg]

Q2 - 2024 (04 Apr Shift 2)

According to Bohr's theory, the moment of momentum of an electron revolving in 4^{th} orbit of hydrogen atom is:

(1) $\frac{h}{\pi}$

(2) $\frac{h}{2\pi}$

(3) $8\frac{h}{\pi}$

(4) $2\frac{h}{\pi}$

Q3 - 2024 (05 Apr Shift 1)

An electron rotates in a circle around a nucleus having positive charge Ze. Correct relation between total energy (E) of electron to its potential energy (U) is :

(1) $\mathrm{E}=\mathrm{U}$

(2) $2\mathrm{E}=\mathrm{U}$

(3) $2\mathrm{E}=3\mathrm{U}$

(4) $\mathrm{E}=2\mathrm{U}$

Q4 - 2024 (05 Apr Shift 2)

The angular momentum of an electron in a hydrogen atom is proportional to :

(Where r is the radius of orbit of electron)

(1) r

(2) $\sqrt{\mathrm{r}}$

(3) $\frac{1}{\sqrt{\mathrm{r}}}$

(4) $\frac{1}{\mathrm{r}}$

Q5 - 2024 (05 Apr Shift 2)

The shortest wavelength of the spectral lines in the Lyman series of hydrogen spectrum is 915Ã.... The longest wavelength of spectral lines in the Balmer series will be _____ Ã....

Q6 - 2024 (06 Apr Shift 1)

The ratio of the shortest wavelength of Balmer series to the shortest wavelength of Lyman series for hydrogen atom is :

(1) 4 : 1

(2) 1 : 4

(3) 2 : 1

(4) 1 : 2

Q7 - 2024 (06 Apr Shift 1)

Radius of a certain orbit of hydrogen atom is 8.48Ã.... If energy of electron in this orbit is E/x. then $x =$ _____

(Given $\mathrm{a}_0 = 0.529$Ã..., $E =$ energy of electron in ground state).

Q8 - 2024 (06 Apr Shift 2)

The longest wavelength associated with Paschen series is : (Given $\mathrm{R_H} = 1.097 \times 10^7$ SI unit)

(1) 3.646×10^{-6} m

(2) 1.876×10^{-6} m

(3) 2.973×10^{-6} m

(4) 1.094×10^{-6} m

Q9 - 2024 (06 Apr Shift 2)

In Franck-Hertz experiment, the first dip in the current-voltage graph for hydrogen is observed at 10.2 V. The wavelength of light emitted by hydrogen atom when excited to the first excitation level is _________ nm. $(\text{Given hc} = 1245 \text{eVnm}, \text{e} = 1.6 \times 10^{-19} \text{C})$.

Q10 - 2024 (08 Apr Shift 1)

In an alpha particle scattering experiment distance of closest approach for the α particle is 4.5×10^{-14} m. If target nucleus has atomic number 80 , then maximum velocity of α - particle is _______ $\times 10^5 \mathrm{~m/s}$ approximately.

$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI unit, mass of } \alpha \text{ particle } = 6.72 \times 10^{-27} \text{ kg}\right)$

Q11 - 2024 (09 Apr Shift 2)

A hydrogen atom in ground state is given an energy of 10.2eV. How many spectral lines will be emitted due to transition of electrons?

(1) 6

(2) 3

(3) 10

(4) 1

Answer Key

Q1 (7) **Q2** (4) **Q3** (2) **Q4** (2)

Q5 (6588) **Q6** (1) **Q7** (16) **Q8** (2)

Q9 (122) **Q10** (156) **Q11** (4)

Q1

$$\Delta E = 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.9\text{eV}$$

$$\Delta E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E}$$

$$P_i = P_t$$

$$0 = -mv + \frac{h}{\lambda''}$$

$$\Rightarrow v = \frac{h}{m\lambda'}$$

$$\Delta E = \frac{1}{2}mv^2 + \frac{hc}{\lambda''}$$

$$= \frac{1}{2}m\left(\frac{h}{m\lambda'}\right)^2 + \frac{hc}{\lambda'}$$

Now

$$\Delta E = \frac{h^2}{2m\lambda'^2} + \frac{hc}{\lambda'}$$

$$\lambda'^2\Delta E - hc\lambda' - \frac{h^2}{2m} = 0$$

$$\lambda' = \frac{hc \pm \sqrt{h^2c^2 + \frac{4\Delta Eh^2}{2m}}}{2\Delta E}$$

$$\lambda' = \frac{hc \pm hc\sqrt{1 + \frac{2\Delta E}{mc}}}{2\Delta E}$$

$$\frac{\lambda'}{\lambda} = \frac{1 + \left(1 + \frac{2\Delta E}{mc^2}\right)^{\frac{1}{2}}}{2} = \frac{1 + 1 + \frac{\Delta E}{mc^2}}{2}$$

$$\frac{\lambda'}{\lambda} = 1 + \frac{\Delta E}{2mc^2}$$

$$\frac{\lambda' - \lambda}{\lambda} = \frac{\Delta E}{2mc^2} = \frac{1.9 \times 1.6 \times 10^{-19}}{2 \times 1.67 \times 10^{-27} \times 9 \times 10^{15}} = 10^{-9}$$

$$\therefore \%\text{ change } \approx 10^{-7}$$

Correct answer 7

Q2

Moment of momentum is $\vec{r} \times \vec{P}$

$\vec{L} = \vec{r} \times m\vec{v}$

$$L = mvr = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$$

Q3

$$F = \frac{k(Ze)(e)}{r^2} = \frac{mv^2}{r}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}\frac{K(Ze)(e)}{r}$$

$$PE = -\frac{K(Ze)(e)}{r}$$

$$TE = \frac{K(Ze)(e)}{2r} - \frac{K(Ze)(e)}{r} = \frac{-K(Ze)(e)}{2r}$$

$$TE = \frac{PE}{2}$$

$$2TE = PE$$

Q4

$$F_C = \frac{mv^2}{r}$$

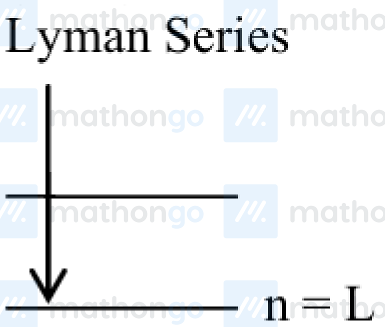
$$\frac{Kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$mv^2r^2 = K\,K_1q_2r$$

$$\frac{L^2}{m} = KKq_1q_2r$$

$$L \propto \sqrt{r}$$

Q5



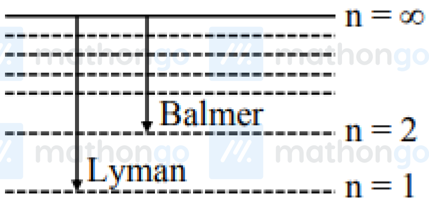
Shortest, $\frac{hc}{\lambda} = -13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\lambda \downarrow E \uparrow; \frac{hc}{\lambda_0} = -13.6(1)$

Balmer Series :

$$\frac{hc}{\lambda_1} = -13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$
$$\frac{hc}{\lambda_1} = -13.6\left(\frac{1}{4} - \frac{1}{9}\right)$$
$$\frac{hc}{\lambda_1} = -13.6 \times \left(\frac{5}{36}\right)$$
$$\Rightarrow \frac{-13.6\lambda_0}{\lambda_1} = -13.6 \times \frac{5}{36}$$
$$\lambda_1 = \frac{\lambda_0 \times 36}{5} = \frac{915 \times 36}{5} = 6588$$

Q6



$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$
$$\frac{\frac{1}{\lambda_L} = Rz^2\left(\frac{1}{1^2}\right)}{\frac{1}{\lambda_B} = Rz^2\left(\frac{1}{2^2}\right)}$$
$$\frac{\lambda_B}{\lambda_L} = 4 : 1$$

Q7

We know

$$r = 0.529\frac{n^2}{Z} \Rightarrow 8.48 = 0.529\frac{n^2}{1}$$
$$n^2 = 16 \Rightarrow n = 4$$

We know

$E \propto \frac{1}{n^2}$

$$E_{n^{\text{th}}} = \frac{E}{16}$$

$x = 16$

Q8

For longest wavelength in Paschen's series:

$\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For longest $n_1 = 3$

$n_2 = 4$

$$\frac{1}{\lambda} = R\left[\frac{1}{(3)^2} - \frac{1}{(4)^2}\right]$$

$$\frac{1}{\lambda} = R\left[\frac{1}{9} - \frac{1}{16}\right]$$

$$\frac{1}{\lambda} = R\left[\frac{16-9}{16 \times 9}\right]$$

$$\Rightarrow \lambda = \frac{16 \times 9}{7R} = \frac{16 \times 9}{7 \times 1.097 \times 10^7}$$

$\lambda = 1.876 \times 10^{-6}$ m

Q9

$$10.2\text{eV} = \frac{hc}{\lambda}$$

$$\lambda = \frac{1245\text{eV} - \text{nm}}{10.2\text{eV}} = 122.06 \text{ nm}$$

Q10

$$v = \sqrt{\frac{4KZe^2}{mr_{\min}}}$$

$$= \sqrt{\frac{4 \times 9 \times 10^9 \times 80}{6.72 \times 10^{-27} \times 4.5 \times 10^{-14}}} \times 1.6 \times 10^{-19}$$

$$= 9.759 \times 10^{25} \times 1.6 \times 10^{-19}$$

$$= 156 \times 10^5 \text{ m/s}$$

Q11

Hydrogen will be in first excited state therefore it will emit one spectral line corresponding to transition b/w energy level 2 to 1